Kartikeya Gangwar

Undergraduate Researcher | Scientific Machine Learning, Geometric DL & Computational Mathematics

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Research Profile

Undergraduate researcher in computational mathematics and scientific machine learning at the University of Delhi. Focus centers on structural geometric invariants, numerical PDEs, and high-performance neural surrogates—including contact state-space models for time series, symplectic potential inversion in astrodynamics (CHRONOS), operator conditioning in incompressible Navier–Stokes flows, and high-dimensional obstacle problems. Author of 10 research manuscripts (including 4 under review at JCP, IEEE TCI, Physics of Fluids, and Emerald IJNMHFF), invited peer reviewer for *Physics of Fluids*, and architect of 2 open-source scientific computing libraries.

Education

2023–2027 **B.Sc. (Honours) Mathematics**, *University of Delhi*, Delhi, India

- **Degree Structure:** 4-Year Undergraduate Research Degree with advanced research specialization.
- **Undergraduate Dissertation Supervisor: Prof. Vinay Kumar** (Department of Mathematics).
- **Relevant Core Coursework:** Real Analysis, Multivariable Calculus, Ordinary & Partial Differential Equations, Complex Analysis, Abstract Algebra, Linear Algebra, Numerical Analysis, Probability & Statistics.
- **Independent Scholarly Trajectory:** Rigorous, self-directed specialization in geometric deep learning, symplectic mechanics, nonlinear PDEs, and high-performance scientific computing developed alongside the formal undergraduate program.

Research Manuscripts & Preprints

Pillar I: Inverse Problems, Symplectic Astrodynamics & Fluid Mechanics

Under Review **Shape Inversion in Electrical Impedance Tomography**, *IEEE Transactions on Computational Imaging (TCI)*

"Derivative-Informed Neural Forward Surrogates with Directional Sensitivity Supervision for Shape Inversion in Electrical Impedance Tomography" **[MS #TCI-04529-2026]**

- **Authors:** **Kartikeya Gangwar** (Sole Author / Lead Investigator).
- **Mathematical Formulation:** Parameterized closed inclusion boundaries via 64-dimensional periodic cubic B-splines coupled to mollified boundary-integral winding indicators. Verified analytic sensitivities $\mathbf{J}_{\mathrm{FEM}} = \mathbf{J}_{\sigma}\frac{\partial \boldsymbol{\Sigma}}{\partial \boldsymbol{\theta}}$ against directional finite differences ($< 3.2 \times 10^{-4}$ discrepancy).
- **Algorithmic Novelty:** Introduced **Stochastic Directional JVP Supervision** on $\mathbb{S}^{63}$, reducing retained tangent autograd branches from 64 to 1 per sample and bounding peak VRAM to $342.8\,\mathrm{MB}$.
- **Empirical Results:** Zero-online-FEM inversion delivers a $\mathbf{56.8}\times$ wall-clock speedup ($35.6\,\mathrm{ms}$ vs $2.02\,\mathrm{s}$/step) with a consensus IoU of 0.8235 and a 99.57% noise robustness rate across 234 trials.
- **Codebase:** `https://github.com/KartikeyaGangwar/eit-neural-surrogate-inversion`

Preprint DOI: 10.5281/zenodo.22096368

Working Paper / Dissertation

CHRONOS: Symplectic Potential Inversion in Celestial Mechanics, *Target: Monthly Notices of the Royal Astronomical Society (MNRAS) / Astronomy & Computing*

"CHRONOS: Canonical Hamiltonian Reconstruction of Orbital Non-Integrable Systems via Symplectic Potential Inversion Networks (SymPIN) Across Autonomous and Non-Autonomous Celestial Regimes" **[Undergraduate Research Dissertation]**

- **Authors:** Kartikeya Gangwar (Lead Author / Primary Investigator), **Prof. Vinay Kumar** (Undergraduate Dissertation Supervisor & Senior Author).
- **Mathematical Formulation:** Proved *Theorem 1* (Separable Symplectic Kinetic-Coriolis Exact Decomposition for Rotating Frames, $\nabla_{\mathbf{z}} \cdot \mathbf{f}_\theta \equiv 0$), *Theorem 2* (Arnold Extended Phase Space $\mathbb{R}^{2d+2}$ with exact energy conservation $\mathcal{K}_\theta \equiv 0$ on Σ_0 for pulsating non-autonomous systems), and *Theorem 3* (Separable Extended Synthesis unifying Coriolis cross-coupling with Jeans-Meshchersky secular mass loss).
- **Empirical Benchmarks Across 7 Systems:** Benchmarked across 7 canonical chaotic celestial mechanical systems (Binary Quasar Plummer, Restricted Six-Body C_{4v}, Photogravitational Chermnykh with Asteroid Belt, Magnetic Binary Yukawa, Elliptic Sitnikov, Variable-Mass Binary, and Pseudo-Newtonian Four-Body with cubic relativistic singularity).
- **Manifold Confinement & Invariant Drift:** Preserved conserved Jacobi/Hamiltonian invariants within symplectic confinement thresholds ($|\Delta\mathcal{I}| < 10^{-1}$ down to 10^{-7}), guaranteeing strictly zero unphysical trajectory departure across chaotic horizons (> 4.37 Lyapunov timescales, $T = 8.0$–30.0).
- **Wavelet Discovery & Ringing Trap Resolution:** Discovered the **Far-Field Oscillatory Ringing Trap** of global Fourier features ($\mathcal{R}_{\text{ring}} = 2.61 \times 10^{10}$); eliminated unphysical micro-ripples via proposed **Mexican Hat Wavelet Net** Poisson dual kernel ($\mathbf{2.0 \times 10^{11}}\times$ cleaner far-field Laplacian curvature, $\mathbf{45.3}\times$ higher near-core resolution).
- **Sundman Symplectic Stepper & Poincaré Mapping:** Formulated adaptive Sundman time regularization ($\mathrm{d}t = g(\mathbf{q})\mathrm{d}s$), resolving periastron encounters with zero secular energy drift (7.98×10^{-6}), generating high-density Poincaré sections ($> 1{,}100$ punctures) resolving KAM tori and chaotic separatrices.
- **Codebase:** `https://github.com/KartikeyaGangwar/cpa-shnn`

Working Paper

Contact-SSM: Geometric State-Space Models for Time Series, *Target: NeurIPS / SIMODS*

"Contact-SSM: Geometric Contact State-Space Models for Long-Horizon Dynamical Time Series Forecasting"

- **Authors:** Kartikeya Gangwar (Lead Investigator / Mathematical Formulation).
- **Formulation:** Embedded latent dynamics on an odd-dimensional contact manifold $\mathbb{R}^{2d+1}$ with contact 1-form $\alpha = \mathrm{d}s - \mathbf{p}^T\mathrm{d}\mathbf{q}$ and Reeb vector field $R = \partial/\partial s$, modeling dissipative and driven dynamical systems without secular phase drift.
- **Parallel Prefix Scan:** Mapped infinitesimal generator $K \in \mathfrak{sp}(2d) \oplus \mathbb{R}$ via Cayley transform into associative scan operators, achieving $\mathcal{O}(L)$ GPU parallel training and $\mathcal{O}(1)$ real-time streaming rollouts.
- **Codebase:** `https://github.com/KartikeyaGangwar/time-series-forecasting`

Under Review

Operator Conditioning & False Convergence in PINNs, *Physics of Fluids (AIP Publishing)*

"Operator Conditioning and False Convergence in Physics-Informed Neural Networks for Incompressible Flows" **[MS #POF26-AR-14856]**

- **Authors:** Kartikeya Gangwar (Sole Author / Lead Investigator).
- **Formulation Investigation:** Conducted controlled formulation study on 2D regularized lid-driven cavity ($Re = 1000$), contrasting Streamfunction–Pressure (ψ–p) against Streamfunction–Vorticity (ψ–ω) across 20,000 Adam + L-BFGS iterations.
- **False Convergence Discovery:** Discovered that ψ–ω PINNs suffer from **False Convergence and Operator Diffusion** ($67.3\times$ residual drop) while underpredicting secondary corner eddy intensity by 14.3% ($\psi_{\min} = 0.845 \times 10^{-3}$) and yielding 14.2% global velocity L_2 error vs 2.8% in ψ–p.
- **Operator Conditioning & Hodge Projection:** Proved continuous mesh-free PINNs lack discrete stencils for Thom's wall-vorticity condition, leaving boundary vorticity under-specified; retaining pressure p in ψ–p enforces an implicit elliptic Helmholtz–Hodge projection that restores operator conditioning.
- **Codebase & Preprint:** `https://github.com/KartikeyaGangwar/pinn-fluid-formulations` **Preprint DOI:** 10.5281/zenodo.22979276

Under Review **Extreme-Re Navier–Stokes Framework & Unsteady Dynamics**, *Int. J. Numer. Meth. Heat & Fluid Flow (Emerald)*

"High-Resolution Finite-Difference Framework for Steady Continuation and Unsteady Dynamics up to $Re = 100{,}000$ in Lid-Driven Cavity Flows" **[MS #HFF-10-2026-1582]**

- **Authors: Kartikeya Gangwar** (Sole Author / Principal Investigator).
- **Exact Spectral Poisson Acceleration:** Formulated an exact $\mathcal{O}(N^2 \log N)$ Discrete Sine Transform (DST-I) Poisson engine with vectorized ADI marching, eliminating iterative residual drift and enabling machine-precision pressure-velocity coupling across 1.05×10^6 nodes (1025×1025).
- **High-Re Frontiers & Batchelor Verification:** Advanced steady continuation up to $Re = 100,000$ on uniform meshes up to 1025^2, observing core vorticity stabilization ($\omega_0 \approx -2.54$, $x_c \to 0.50000$) aligned with Batchelor's core theorem; verified numerical uncertainty via ASME V&V 20 Richardson GCI ($\text{GCI}_{32} = 0.48\%$).
- **Directional Hybrid Regularization:** Maintained strictly zero artificial viscosity ($\nu_{\text{art}} = 0$) for all $Re \leq 50,000$ via pure second-order central differencing; regularized corner singularities at $Re = 100,000$ ($Pe = 195.3$) via directional hybrid differencing with explicit numerical viscosity tensor mapping.
- **Finite-Window Unsteady Marching:** Executed time-accurate marching on 513×513 meshes ($t = 16.0$–22.0 c.t.u., $St \equiv fL/U$), resolving coherent vortex shedding ($Re = 10,000$, $St = 0.6249 \pm 0.0416$), amplitude-modulated dynamics ($Re = 15,000$, $St = 0.5364$), and finite-window spectral responses up to $Re = 100,000$.
- **Open-Source Benchmarks & Datasets:** Released 30 verified flow-field datasets with SHA-256 integrity on Zenodo and GitHub: `https://github.com/KartikeyaGangwar/lid-driven-cavity-cfd` **DOI:** 10.5281/zenodo.18312938

Pillar II: The Adaptive Subspace (AS) Optimization Paradigm

Under Review **Parameter Decoupling, Conditioning & Interface Transmission**, *Journal of Computational Physics (JCP, Elsevier)*

"On Parameter Decoupling, Conditioning, and Interface Transmission in Multi-Objective Physics-Informed Neural Networks" **[MS #JCOMP-D-26-03600]**

- **Authors: Kartikeya Gangwar** (First & Corresponding Author), **Samarjeet Malik** (Co-Author).
- **Mathematical Formulation:** Constructed an algebraic direct-sum decomposition ($\Theta = \Theta_0 \oplus \Theta_1, \mathcal{W}_0 \mathcal{W}_1^T = \mathbf{0}$) with C^2 **Quintic Hermite routing** ($\psi(\xi) = 6\xi^5 - 15\xi^4 + 10\xi^3$), proving exact gradient orthogonality ($\langle \nabla \mathcal{L}_{\text{if}}, \nabla \mathcal{L}_{\text{des}} \rangle \equiv 0$) on saturated domains with zero jump discontinuities ($\nabla^2 u \in C^0$).
- **Theoretical & Duality Discoveries:** Proved representation rank invariance ($\text{rank}(\mathbf{H}_\alpha \mathbf{H}_\alpha^T) = 2, \forall \alpha < 1$) establishing continuous sharing as an optimizer preconditioner. Using Green's representation theory, proved that compact cubic B-spline bases resolve boundary stiffness ($1.53\times$ lower interior error, 2.08×10^{-4} interface loss) without parameter splitting.
- **Multi-Physics Verification:** Evaluated across 10 matched-capacity models ($P = 9{,}216$, 8 seeds) and 6 physical regimes, including 3D Linear Paul Ion Traps ($2.0\times$ harmonic purity), high-contrast iron magnets, and non-linear phase-field/shock systems.
- **Codebase & Preprint:** `https://github.com/KartikeyaGangwar/null-space-pinn` **Preprint DOI:** 10.5281/zenodo.22132799

Preprint **Adaptive N-Subspace PINN (AS-PINN)**, *Target: Journal of Computational Physics (JCP) / SISC*

"Adaptive N-Subspace Physics-Informed Neural Networks (AS-PINN): Autonomous Domain Decomposition via Parameter-Space Adaptive Mesh Refinement"

- **Authors: Kartikeya Gangwar** (Sole Author / Principal Investigator).
- **Algorithmic Novelty:** Developed an autonomous parameter-space AMR engine using vectorized per-sample Gram alignment profiling ($\mathcal{G}_{ij} = \cos\angle(\mathbf{g}_i, \mathbf{g}_j)$ via `torch.func.vmap`) with exact zero-disruption cleavage invariance ($\|u^{(N+1)} - u^{(N)}\| = 0$).
- **Empirical Verification:** Validated across 9 canonical PDEs (Burgers, Convection 2D, Quantum NLS, 4D Diffusion, Bratu 2D, Kovasznay, Cavity NS, Klein-Gordon), achieving a $\mathbf{725.6}\times$ loss reduction over standard PINNs, PCGrad, and CAGrad on high-frequency Helmholtz ($k = 4\pi$).
- **Codebase & Preprint:** `https://github.com/KartikeyaGangwar/as-pinn` **Preprint DOI:** 10.5281/zenodo.22822521

Preprint **AS-ViT: Adaptive Subspace Vision Transformers**, *Target: IEEE TPAMI / CVPR*

"AS-ViT: Adaptive Subspace Vision Transformers with Autonomous Gradient-Clash Routing for Multi-Task Learning"

- **Authors: Kartikeya Gangwar** (Sole Author / Principal Investigator).
- **Architectural Innovation:** Resolved negative transfer in dense multi-task vision backbones (segmentation, depth, normals, edges) by continuously tracking inter-task Gram matrix eigenvalues ($\lambda_{\min}(\mathcal{G}) < -\tau_{\text{conflict}}$) and cleaving dedicated latent expert subspaces using Partition of Unity (PoU) gating.
- **Empirical Results:** Outperformed monolithic ViTs, PCGrad, CAGrad, and fixed MoEs on NYUv2, Cityscapes, and Taskonomy benchmarks, achieving $+5.84\%$ mean multi-task gain (ΔM) with zero manual expert tuning.
- **Codebase & Preprint:** `https://github.com/KartikeyaGangwar/as-vit-multitask` **Preprint DOI:** 10.5281/zenodo.22974130

Pillar III: High-Dimensional Stochastic Systems & Uncertainty Quantification

Working Paper **High-Dimensional American Basket Options (Deep-EEP-PINN)**, *Target: Quantitative Finance*

"Deep Early Exercise Premium Physics-Informed Neural Network for High-Dimensional Free-Boundary Obstacle Problems"

- **Authors: Kartikeya Gangwar** (Lead Author / Principal Investigator).
- **Mathematical Formulation:** Embedded the analytical Kim (1990) / Carr-Jarrow-Myneni (1992) Early Exercise Premium decomposition into a hard-constrained boundary ansatz: $V_{\text{Amer}} = V_{\text{Euro}}^{\text{exact}} + e_\theta(\mathbf{S}, t)$ with $e_\theta(\mathbf{S}, T) \equiv 0$.
- **High-Dimensional Breakthrough:** Priced correlated American basket options up to $\mathbf{d = 50}$ **assets (1,225 correlations)**, breaking Bellman's Curse of Dimensionality (N^d) via Directional Autograd Hessian Trace Contraction in $\mathcal{O}(d)$ linear complexity ($< 3\,\text{GB}$ VRAM). Validated against 100,000-path Longstaff-Schwartz Monte Carlo and Crank-Nicolson PSOR (0.37% relative error).
- **Codebase:** `https://github.com/KartikeyaGangwar/Deep-EEP-PINN`

Working Paper **Bayesian PINN Fidelity & Posterior Distortion**, *Target: Reliability Engineering & System Safety / Inverse Problems*

"Bayesian PINN Fidelity & Parameter-Resolved Error Localization Framework"

- **Authors: Kartikeya Gangwar** (Lead Author / Principal Investigator).
- **Epistemological Investigation:** Formally investigated how PINN forward surrogate approximation errors distort Bayesian inverse posteriors.
- **Proposed Diagnostic Metric:** Formulated the **Bayesian Fidelity Ratio (BFR)**: $\text{BFR} = \mathcal{W}_1(\pi_E, \pi_A)/\mathcal{W}_1(\pi_{E,1}, \pi_{E,2})$ based on 1-Wasserstein optimal transport distance normalized by empirical MCMC stochastic noise floors under paired control chains.
- **Codebase:** `https://github.com/KartikeyaGangwar/PINN-Bayesian-Posterior-Fidelity`

Scientific Software & Computational Infrastructure

Pillar IV: Sim2Real Autonomous Vehicle Dynamics & Scientific Infrastructure

Preprint / v2.0.1 **BeamNG Research SDK (BRSDK)**, *Target: JOSS (2027) / Open-Source*

"BRSDK: A High-Frequency In-Engine Telemetry Framework and Python SDK for Sim2Real Autonomous Vehicle Research in BeamNG"

- **Architect & Lead Developer: Kartikeya Gangwar** (Sole Author / Lead Architect).
- **Sim2Real Physics Instrumentation:** Formulated an in-engine extraction architecture operating in BeamNG's **2,000 Hz soft-body physics thread** to resolve sub-millisecond transient dynamics (tire slip breakaway energy, dynamic damper velocities, and chassis compliance) necessary for robust simulation-to-real transfer.
- **Zero-Allocation Memory Design:** Engineered a dual-runtime pipeline eliminating dynamic Lua heap allocations during logging, preserving deterministic simulation stepping and continuous ground-truth physical state trajectories without garbage collection pauses.
- **Scientific Python SDK & ML Interface:** Built an Apache Arrow columnar Python SDK (`brsdk`) powered by Polars, featuring strict Pydantic V2 metadata sidecar boundaries and zero-copy DLPack interchange with PyTorch and Minari for offline reinforcement learning.
- **Codebase & Dissemination:** `https://github.com/KartikeyaGangwar/BRSDK` **DOI:** 10.5281/zenodo.21729606 **Release:** `v2.0.1`

Open-Source **Extreme-Re Navier–Stokes Suite & Benchmark Engine**, *Python / NumPy / SciPy*

- **Lead Developer: Kartikeya Gangwar**.
- **Spectral Poisson Engine:** Integrated an exact $\mathcal{O}(N^2 \log N)$ Discrete Sine Transform (DST-I) solver, eliminating iterative Poisson residuals and enabling machine-precision pressure-velocity coupling across 1.05×10^6 nodes (1025×1025).
- **Continuation & Regularization:** Engineered continuation across 23 Reynolds numbers ($Re = 100 \rightarrow 100,000$), preserving pure central differencing ($\nu_{\text{art}} = 0$) for $Re \leq 50,000$ and applying directional hybrid differencing at $Re = 100,000$.
- **Verification & Open Dissemination:** Fully automated ASME V&V 20 Richardson GCI verification, real-time unsteady probe telemetry, and animated shedding pipelines; released 30 verified flow-field datasets.
- **Open-Source Dissemination:** `https://github.com/KartikeyaGangwar/lid-driven-cavity-cfd` **DOI:** 10.5281/zenodo.18312938

Technical Expertise & Methodologies

Scientific Machine Learning: Physics-Informed Neural Networks (PINNs), Hamiltonian Neural Networks (HNNs), Symplectic Neural ODEs, Deep Surrogate Inversion, Boundary-Encoded Neural Ansätze, Wavelet Encodings (Mexican Hat / Morlet), SymPIN Inversion, Fourier Encodings, Neural Operators.

Theoretical Mathematics: Symplectic Geometry & Differential Forms, Canonical Phase Space Topologies, Contact Hamiltonian Mechanics, Sundman Regularization, Nonlinear Parabolic & Elliptic PDEs, Free-Boundary Obstacle Problems, Optimal Transport & Wasserstein Metrics, Functional Analysis, Operator Theory.

Autograd & Deep Learning Engineering: PyTorch (low-level custom autograd engines, `torch.func.vmap`, forward-mode automatic differentiation, Jacobian-Vector Products (JVP), Hessian-vector contractions), CUDA acceleration, GPU memory profiling, mixed-precision arithmetic.

Numerical Analysis & Optimization: Finite Element Method (FEM), Finite Difference Schemes (ADI, Red-Black SOR, Crank-Nicolson), Symplectic Runge-Kutta (RK4), L-BFGS (Strong-Wolfe curvature line search), AdamW, Gradient Surgery (PCGrad, CAGrad), Markov Chain Monte Carlo (MCMC, Metropolis-Hastings).

Languages & Developer Toolchain: Python (NumPy, SciPy, Matplotlib, PyTorch), C++, Git/GitHub, Linux/Bash, LaTeX, VS Code, Kaggle GPU Environments.

Academic Service & Peer Review

2026 **Invited Peer Reviewer**, *Physics of Fluids (AIP Publishing)*
Official peer reviewer for research manuscript on vortex dynamics and vorticity-transport mechanisms [MS #POF26-AR-13153R].

Scholarly Philosophy

Open Science & Rigor: Firm conviction in reproducible computational science. Every research framework is accompanied by self-contained automated benchmark runners, zero-redundancy unit testing, full ablation matrices, and permissive open-source licensing (MIT / Apache 2.0).